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LLM SPOKEN CONVERSATION

REVISITING AND EXTENDING SPOKEN CONVERSATION SIMULATION

TECHNION Israel Institute of Technology | DATA AND DECISION SCIENCES MAKING SENSE OF IT

ORIGINAL STUDY

- Mayor et al. investigated whether LLMs can simulate natural spoken telephone conversations.
- Generated LLM conversations were compared with human Switchboard conversations.

RESEARCH QUESTION

How do generation architecture and prompt strategy affect how closely LLM conversations resemble real human telephone conversations?

EXPERIMENTAL DESIGN

4 GENERATION ARCHITECTURES × 3 PROMPTS = 12 CONDITIONS
THEN : 12 CONDITIONS X 50 CONVISRATIONS= 600 CONVERSATIONS
MODELS: VICUNA-13B (ALL ARCHITECTURES) + MISTRAL-7B (SECOND AGENT IN A4)
HUMAN REFERENCE: SWITCHBOARD TELEPHONE CALLS

C1 : All-at-once

C2: Single model turn-by-turn

A3: Independent same-model agents

A4: Independent cross-model agents

P0 : Basic (Baseline)

P1: Spoken style+persona (No marker forcing)

P2: P1 + real Switchboard example

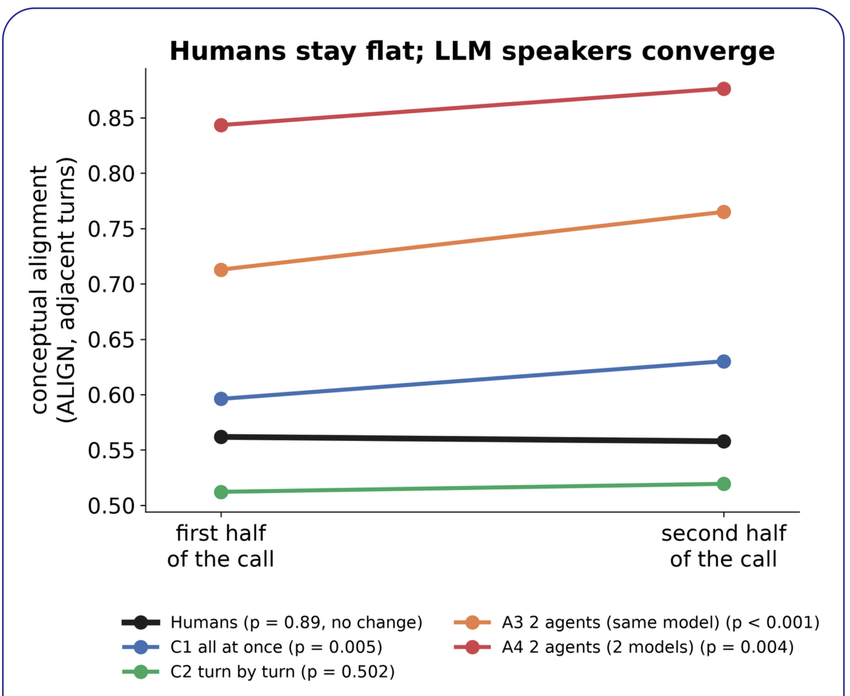
Why we revisited it ?

- In the original setup, a single model generated both sides of the conversation
- Some prompts explicitly encouraged spoken markers and conversation length
- We therefore designed a cleaner comparison between generation architecture and prompt strategy.

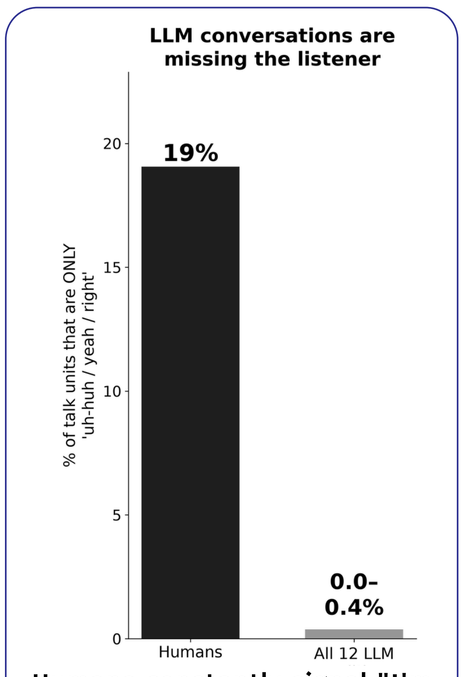
METRICS

Turn Length: Median and Mean to capture typical turn size.
Backchannels: Overall rate, standalone listener turns, and expression diversity.
Dialogue Structure: Dialogue-act distribution and Jensen–Shannon distance from human Switchboard conversations.
Conceptual Alignment: Semantic similarity between adjacent turns using ALIGN.

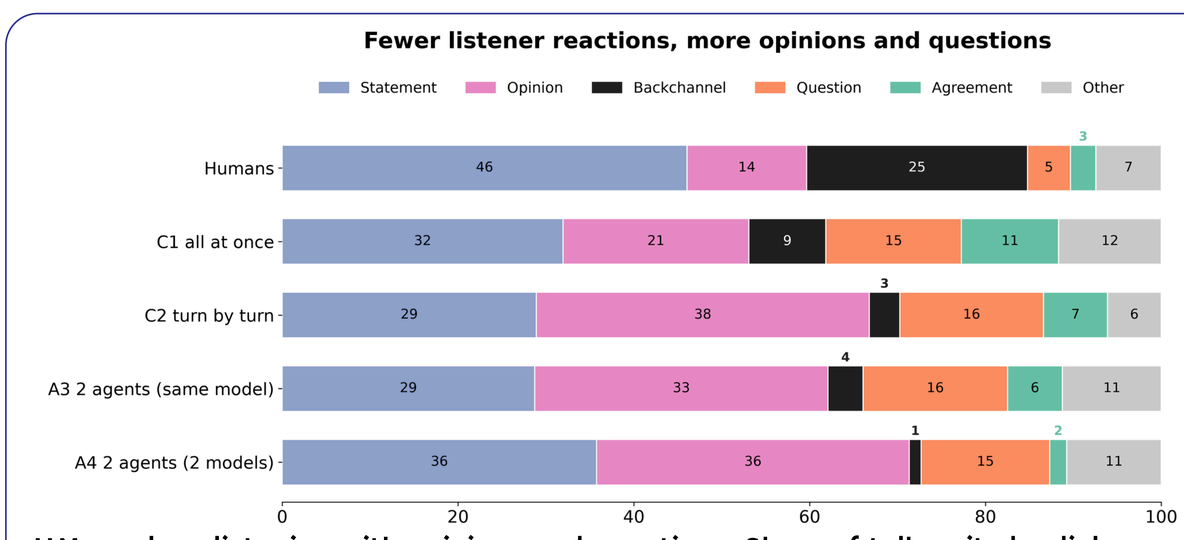
MAIN FINDINGS



Alignment between neighbouring turns, first vs second half of the call (prompt PI)



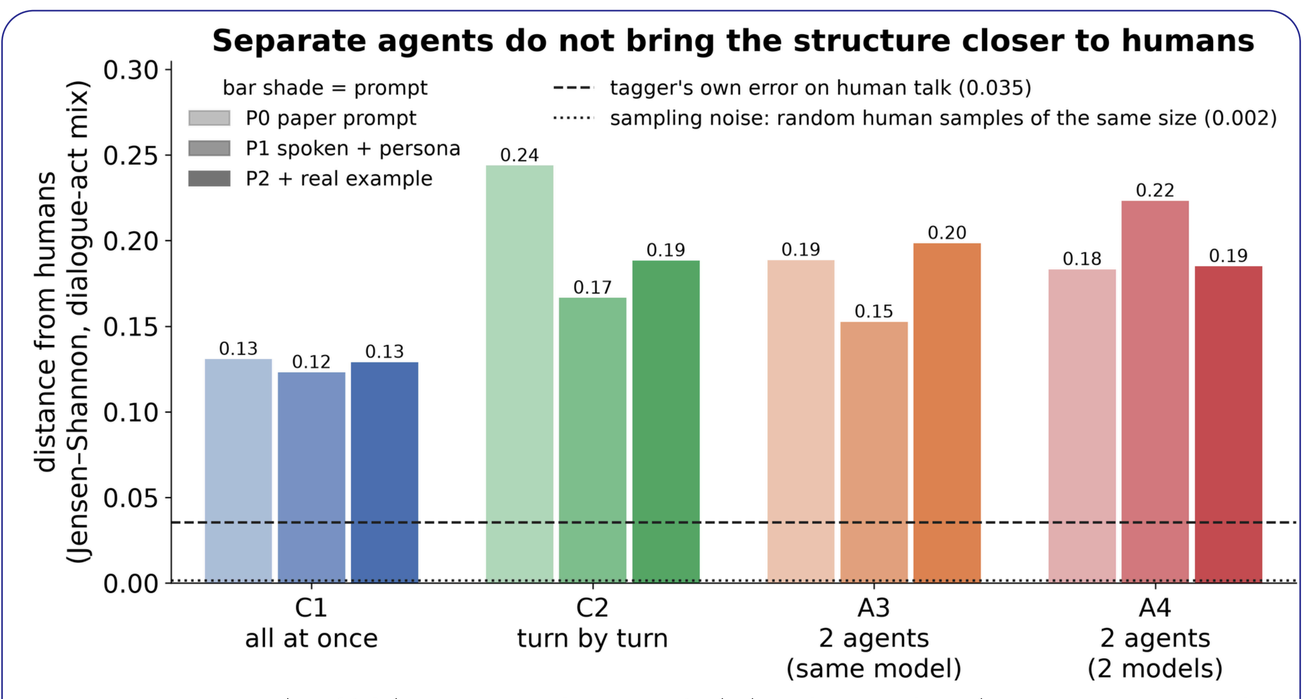
Humans constantly signal "I'm listening"; LLMs almost never do, in any architecture or prompt.



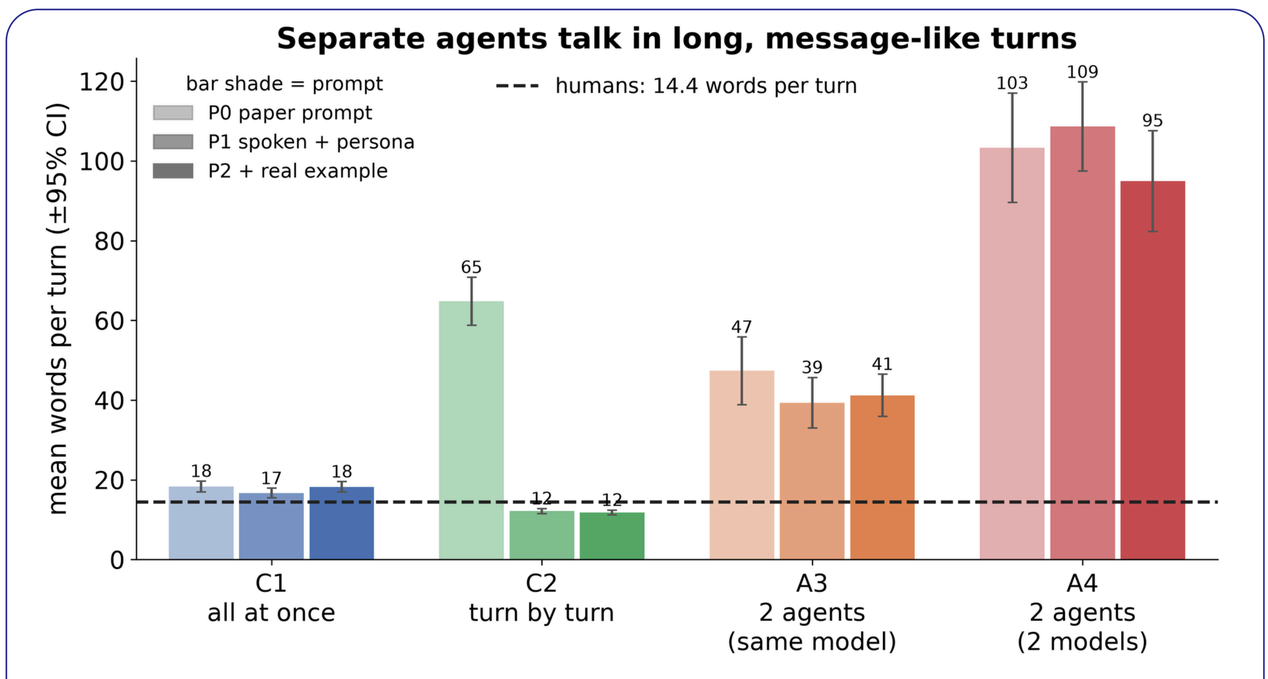
LLMs replace listening with opinions and questions. Share of talk units by dialogue act (automatic tagger, same for humans and LLMs). LLM rows are averaged over prompts P0–P2.

Can we trust the results?

- Tagger checked by hand: 4 annotators labelled 100 LLM sentences. The tagger matched them 77% of the time (annotators agreed with each other 83%).



All-at-once generation (CI) is closest to humans. Splitting the speakers into agents moves the structure further away. Every gap is far above measurement error.



Agent architectures write messages, not spoken turns: up to 7× longer than human turns.

LIMITATIONS

- Limited model coverage : Only two open-source model families were tested.
- Limited sample size : 50 conversations per condition.
- Methodological limitations : Strict turn-taking differs from natural conversation; automatic dialogue-act tagging introduces measurement error.

DISCUSSION

- Giving each speaker its own agent did not make conversations more human-like.
- Architecture had a stronger effect than prompting,while the missing listener role remained a major gap.